Kieran Shah McDaniel

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Education

Harvard University Cambridge, MA
Bachelor of Science (B.S.) in Electrical Engineering | GPA: 4.0 May 2027
Awards: Best Poster at Harvard CCB Symposium 2025, Detur Prize, John Harvard Scholarship

Technical Skills

- C++, Python, MATLAB; embedded systems and firmware (ESP32, nRF52840, BLE, I²C, SPI); PCB design (KiCad, Eagle); LTspice; feedback control; sensor integration; CAD and rapid prototyping (Fusion 360, Rhino); benchtop validation

Projects

Optical to Electrical Receiver | Hardware Designer **May 2026 - present | Cambridge, MA**
- Designed two custom PCBs of a common-emitter amplifier for GHz-bandwidth receiver circuitry targeting data center communication systems.
- Matched transistor linear region to photodiode output and reduced power consumption by 3x without sacrificing linearity.
- Contributed to the tapeout of a mixed-signal integrated circuit for optical-to-electrical signal conversion.
- Designed a custom validation PCB with eight DAC channels, including power delivery, clocking architecture, and an XEM Series FPGA interface.

Tactile Sensing System for Precision Robotics | Researcher **March 2026 - present | Cambridge, MA**
- Designed a custom linear actuator using largely 3D printed components to apply calibrated forces to photo-elastic polymers for tactile sensing.
- Engineered a snap-fit prototype enclosure and integrated optical sensing hardware.
- Iteratively experimented to optimize force sensitivity and spatial resolution of the system.

Humanitarian Field Digital Tally Counter| Electrical Team Lead **August 2025 - December 2025 | Cambridge, MA**
- Led a five-person electrical engineering team from high-level design through a functional handheld prototype as part of an engineering design course.
- Reduced GPIO pin usage by 50% using I²C and SPI protocols and integrated haptic feedback, LEDs, and microSD storage .
- Developed low-power embedded firmware in C++ enabling BLE communication with 8+ hours of battery life.
- Presented weekly technical design reviews and engineering milestones to the Harvard Humanitarian Initiative.

Polargraph Machine | Hardware Designer **February 2024 - September 2025 | Cambridge, MA**
- Designed a custom sliding chassis in Fusion 360 and implemented repeatable positioning using limit switches.
- Reduced low-speed stepper resonance by 70% by implementing microstepping and adjusting acceleration profiles.
- Cut 100g from end-effector by replacing an actively actuated pen-lift with a spring-assisted gravity drop and a smaller reset servo, improving cornering ability.

Professional Experience

Until Labs | Engineering Intern **June 2025 - August 2025 | San Francisco, CA**
- Optimized high-power H-bridge circuitry to reduce switching losses and improve power conversion efficiency.
- Engineered magnetic nanoparticles for rewarming of cryopreserved tissue by tuning cluster and core size to maximize specific loss power while preserving biocompatibility.
- Collaborated across electrical engineering, chemistry, and bioengineering disciplines to rapidly prototype and validate experimental designs.

Engineering Systems and Control | Teaching Assistant **May, 2026 - present | Cambridge, MA**
- Designed hands-on laboratory curriculums to reinforce feedback control principles.
- Mentored students through hardware debugging and circuit validation using oscilloscopes, multimeters, power supplies, and function generators.
- Led weekly office hours to help students develop intuition for analog electronics, MATLAB, and control theory through one-on-one instruction.

The RoboHub | Lead Teacher **May 2026 - present | Cambridge, MA**
- Developed a new electrical engineering curriculum covering circuit analysis, sensors, actuators, and embedded systems.
- Mentored student teams competing in the international VEX IQ Level Up robotics competition.